

# Packages

# Are you struggling?

How people think programming looks



How it really looks 90% of the time



# Plan for today

- Brief recap
- **Packages**

# Recap of using variables

- Most of the time, we want to do more than add, subtract, multiply etc.
- We want to act on our variables. We do this with operators & **functions**
  - *Each function* has a unique name
  - *Each function* requires some input, and the function can be modified using arguments
  - *Each function* will produce an output
- Remember:
  - objects are *nouns*
  - functions & operators are *verbs*
  - arguments are *adverbs*

# Recap of using variables

Where do you find functions?

- Some exist in R by default
  - `t.test()`
  - `cor()`
  - `scale()`

Lots of people all over the world write their own functions. And they (rightly!) think it's useful to share these functions.

# Packages

What is a **package**?

- A collection of functions and datasets
- Open source

Packages are the reason R is so powerful!

- While you can definitely write your own functions, most of what you need to do someone else has already done for you!

| Files Plots <b>Packages</b> Help Viewer                                                                                                                                                                                                                                                                                                                                               |                            |                                                                   |          |                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Install  Update  Packrat <input type="text"/>  |                            |                                                                   |          |                                                                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                       | Name                       | Description                                                       | Ver...   |                                                                                                                                                                             |
| <b>System Library</b>                                                                                                                                                                                                                                                                                                                                                                 |                            |                                                                   |          |                                                                                                                                                                             |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">abind</a>      | Combine Multidimensional Arrays                                   | 1.4-5    |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">acepack</a>    | ACE and AVAS for Selecting Multiple Regression Transformations    | 1.4.1    |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">arm</a>        | Data Analysis Using Regression and Multilevel/Hierarchical Models | 1.11-1   |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">AsioHea...</a> | 'Asio' C++ Header Files                                           | 1.12.2-1 |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">askpass</a>    | Safe Password Entry for R, Git, and SSH                           | 1.1      |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">assertthat</a> | Easy Pre and Post Assertions                                      | 0.2.1    |     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">backports</a>  | Reimplementations of Functions Introduced Since R-3.0.0           | 1.1.6    |   |
| <input checked="" type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                   | <a href="#">base</a>       | The R Base Package                                                | 4.0.0    |                                                                                                                                                                             |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                              | <a href="#">base64...</a>  | Tools for base64 encoding                                         | 0.1-3    |   |

| Files                                                                                     | Plots      | Packages                                                                                 | Help    | Viewer                                                                                    |                                                                                       |                                                                                     |
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|  Install |            |  Update |         |  Packrat |                                                                                       |  |
|                                                                                           | Name       | Description                                                                              | Ver...  |                                                                                           |                                                                                       |                                                                                     |
| <b>System Library</b>                                                                     |            |                                                                                          |         |                                                                                           |                                                                                       |                                                                                     |
| <input type="checkbox"/>                                                                  | abind      | Combine Multidimensional Arrays                                                          | 1.4-5   |        |    |                                                                                     |
| <input type="checkbox"/>                                                                  | acepack    | ACE and AVAS for Selecting Multiple Regression Transformations                           | 1.4.1   |        |    |                                                                                     |
| <input type="checkbox"/>                                                                  | arm        | Data Analysis Using Regression and Multilevel/Hierarchical Models                        | 1.11-1  |        |    |                                                                                     |
| <input type="checkbox"/>                                                                  | AsioHea... | 'Asio' C++ Header Files                                                                  | 1.12.21 |        |    |                                                                                     |
| <input type="checkbox"/>                                                                  | askpass    | Safe Password Entry for R, Git, and SSH                                                  | 1.1     |        |    |                                                                                     |
| <input type="checkbox"/>                                                                  | assertthat | Easy Pre and Post Assertions                                                             | 0.2.1   |       |   |                                                                                     |
| <input type="checkbox"/>                                                                  | backports  | Reimplementations of Functions Introduced Since R-3.0.0                                  | 1.1.6   |      |  |                                                                                     |
| <input checked="" type="checkbox"/>                                                       | base       | The R Base Package                                                                       | 4.0.0   |                                                                                           |                                                                                       |                                                                                     |
| <input type="checkbox"/>                                                                  | base64...  | Tools for base64 encoding                                                                | 0.1-3   |      |  |                                                                                     |

| Files                                                                                     | Plots                                                                                    | Packages                                                                                  | Help                 | Viewer                                                                                |                                                                                       |  |
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| <b>System Library</b>                                                                     |                                                                                          |                                                                                           |                      |                                                                                       |                                                                                       |  |
| <input type="checkbox"/>                                                                  | abind                                                                                    | Combine Multidimensional Arrays                                                           | 1.4-5                |    |    |  |
| <input type="checkbox"/>                                                                  | acepack                                                                                  | ACE and AVAS for Selecting Multiple Regression Transformations                            | 1.4.1                |    |    |  |
| <input type="checkbox"/>                                                                  | arm                                                                                      | Data Analysis Using Regression and Multilevel/Hierarchical Models                         | 1.11-1               |    |    |  |
| <input type="checkbox"/>                                                                  | AsioHea...                                                                               | 'Asio' C++ Header Files                                                                   | 1.12.21              |    |    |  |
| <input type="checkbox"/>                                                                  | askpass                                                                                  | Safe Password Entry for R, Git, and SSH                                                   | 1.1                  |    |    |  |
| <input type="checkbox"/>                                                                  | assertthat                                                                               | Easy Pre and Post Assertions                                                              | 0.2.1                |   |   |  |
| <input type="checkbox"/>                                                                  | backports                                                                                | Reimplementations of Functions Introduced Since R-3.0.0                                   | 1.1.6                |  |  |  |
| <input checked="" type="checkbox"/>                                                       | base                                                                                     | The R Base Package                                                                        | 4.0.0                |                                                                                       |                                                                                       |  |
| <input type="checkbox"/>                                                                  | base64...                                                                                | Tools for base64 encoding                                                                 | 0.1-3                |  |  |  |

| Files                                                                                     | Plots                                                                                    | Packages                                                                                  | Help                                                                                  | Viewer                                                                                |                                                                                     |  |
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| <b>System Library</b>                                                                     |                                                                                          |                                                                                           |                                                                                       |                                                                                       |                                                                                     |  |
| <input type="checkbox"/> <a href="#">abind</a>                                            | Combine Multidimensional Arrays                                                          | 1.4-5                                                                                     |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">acepack</a>                                          | ACE and AVAS for Selecting Multiple Regression Transformations                           | 1.4.1                                                                                     |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">arm</a>                                              | Data Analysis Using Regression and Multilevel/Hierarchical Models                        | 1.11-1                                                                                    |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">AsioHea...</a>                                       | 'Asio' C++ Header Files                                                                  | 1.12.2-1                                                                                  |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">askpass</a>                                          | Safe Password Entry for R, Git, and SSH                                                  | 1.1                                                                                       |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">assertthat</a>                                       | Easy Pre and Post Assertions                                                             | 0.2.1                                                                                     |   |   |                                                                                     |  |
| <input type="checkbox"/> <a href="#">backports</a>                                        | Reimplementations of Functions Introduced Since R-3.0.0                                  | 1.1.6                                                                                     |  |  |                                                                                     |  |
| <input checked="" type="checkbox"/> <a href="#">base</a>                                  | The R Base Package                                                                       | 4.0.0                                                                                     |                                                                                       |                                                                                       |                                                                                     |  |
| <input type="checkbox"/> <a href="#">base64...</a>                                        | Tools for base64 encoding                                                                | 0.1-3                                                                                     |  |  |                                                                                     |  |

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| <b>System Library</b>                                                                     |                                                                                          |                                                                                           |                      |                                                                                       |                                                                                       |                                                                                     |  |
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| <input type="checkbox"/>                                                                  | arm                                                                                      | Data Analysis Using Regression and Multilevel/Hierarchical Models                         | 1.11-1               |    |    |                                                                                     |  |
| <input type="checkbox"/>                                                                  | AsioHea...                                                                               | 'Asio' C++ Header Files                                                                   | 1.12.2-1             |    |    |                                                                                     |  |
| <input type="checkbox"/>                                                                  | askpass                                                                                  | Safe Password Entry for R, Git, and SSH                                                   | 1.1                  |    |    |                                                                                     |  |
| <input type="checkbox"/>                                                                  | assertthat                                                                               | Easy Pre and Post Assertions                                                              | 0.2.1                |   |   |                                                                                     |  |
| <input type="checkbox"/>                                                                  | backports                                                                                | Reimplementations of Functions Introduced Since R-3.0.0                                   | 1.1.6                |  |  |                                                                                     |  |
| <input checked="" type="checkbox"/>                                                       | base                                                                                     | The R Base Package                                                                        | 4.0.0                |                                                                                       |                                                                                       |                                                                                     |  |
| <input type="checkbox"/>                                                                  | base64...                                                                                | Tools for base64 encoding                                                                 | 0.1-3                |  |  |                                                                                     |  |

Files Plots Packages Help Viewer

 Install  Update  Packrat

|                                     | Name                       | Description                                                       | Ver...  |                                                                                       |                                                                                       |
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| <b>System Library</b>               |                            |                                                                   |         |                                                                                       |                                                                                       |
| <input type="checkbox"/>            | <a href="#">abind</a>      | Combine Multidimensional Arrays                                   | 1.4-5   |    |    |
| <input type="checkbox"/>            | <a href="#">acepack</a>    | ACE and AVAS for Selecting Multiple Regression Transformations    | 1.4.1   |    |    |
| <input type="checkbox"/>            | <a href="#">arm</a>        | Data Analysis Using Regression and Multilevel/Hierarchical Models | 1.11-1  |    |    |
| <input type="checkbox"/>            | <a href="#">AsioHea...</a> | 'Asio' C++ Header Files                                           | 1.12.21 |    |    |
| <input type="checkbox"/>            | <a href="#">askpass</a>    | Safe Password Entry for R, Git, and SSH                           | 1.1     |    |    |
| <input type="checkbox"/>            | <a href="#">assertthat</a> | Easy Pre and Post Assertions                                      | 0.2.1   |   |   |
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| <input checked="" type="checkbox"/> | <a href="#">base</a>       | The R Base Package                                                | 4.0.0   |                                                                                       |                                                                                       |
| <input type="checkbox"/>            | <a href="#">base64...</a>  | Tools for base64 encoding                                         | 0.1-3   |  |  |

| Files                                                                                     | Plots                                                                                    | Packages                                                                                  | Help                                                                                  | Viewer                                                                                |                                                                                     |  |
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| <input type="checkbox"/> <a href="#">acepack</a>                                          | ACE and AVAS for Selecting Multiple Regression Transformations                           | 1.4.1                                                                                     |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">arm</a>                                              | Data Analysis Using Regression and Multilevel/Hierarchical Models                        | 1.11-1                                                                                    |    |    |                                                                                     |  |
| <input type="checkbox"/> <a href="#">AsioHea...</a>                                       | 'Asio' C++ Header Files                                                                  | 1.12.2-1                                                                                  |    |    |                                                                                     |  |
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| <input checked="" type="checkbox"/> <a href="#">base</a>                                  | The R Base Package                                                                       | 4.0.0                                                                                     |                                                                                       |                                                                                       |                                                                                     |  |
| <input type="checkbox"/> <a href="#">base64...</a>                                        | Tools for base64 encoding                                                                | 0.1-3                                                                                     |  |  |                                                                                     |  |

| Files                                                                                     | Plots                                                                                    | Packages                                                                                  | Help                 | Viewer                                                                                |                                                                                       |  |
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| <b>System Library</b>                                                                     |                                                                                          |                                                                                           |                      |                                                                                       |                                                                                       |  |
| <input type="checkbox"/>                                                                  | abind                                                                                    | Combine Multidimensional Arrays                                                           | 1.4-5                |    |    |  |
| <input type="checkbox"/>                                                                  | acepack                                                                                  | ACE and AVAS for Selecting Multiple Regression Transformations                            | 1.4.1                |    |    |  |
| <input type="checkbox"/>                                                                  | arm                                                                                      | Data Analysis Using Regression and Multilevel/Hierarchical Models                         | 1.11-1               |    |    |  |
| <input type="checkbox"/>                                                                  | AsioHea...                                                                               | 'Asio' C++ Header Files                                                                   | 1.12.2-1             |    |    |  |
| <input type="checkbox"/>                                                                  | askpass                                                                                  | Safe Password Entry for R, Git, and SSH                                                   | 1.1                  |    |    |  |
| <input type="checkbox"/>                                                                  | assertthat                                                                               | Easy Pre and Post Assertions                                                              | 0.2.1                |   |   |  |
| <input type="checkbox"/>                                                                  | backports                                                                                | Reimplementations of Functions Introduced Since R-3.0.0                                   | 1.1.6                |  |  |  |
| <input checked="" type="checkbox"/>                                                       | base                                                                                     | The R Base Package                                                                        | 4.0.0                |                                                                                       |                                                                                       |  |
| <input type="checkbox"/>                                                                  | base64...                                                                                | Tools for base64 encoding                                                                 | 0.1-3                |  |  |  |

# How do I get packages?

Packages can be downloaded from the *CRAN* (Comprehensive *R* Archive Network)

You will do this from inside [R](#)

**Need to be connected to the internet!**

# 2 ways to install packages

1. Install button in the Packages tab
2. R Code

*Either way, you need to know the name of the package*

# Install Button

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN)

Packages (separate multiple with space or comma):

Install to Library:

/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C]

☒ Install dependencies

Install Cancel

# Install Button

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN)

Packages (separate multiple with space or comma):

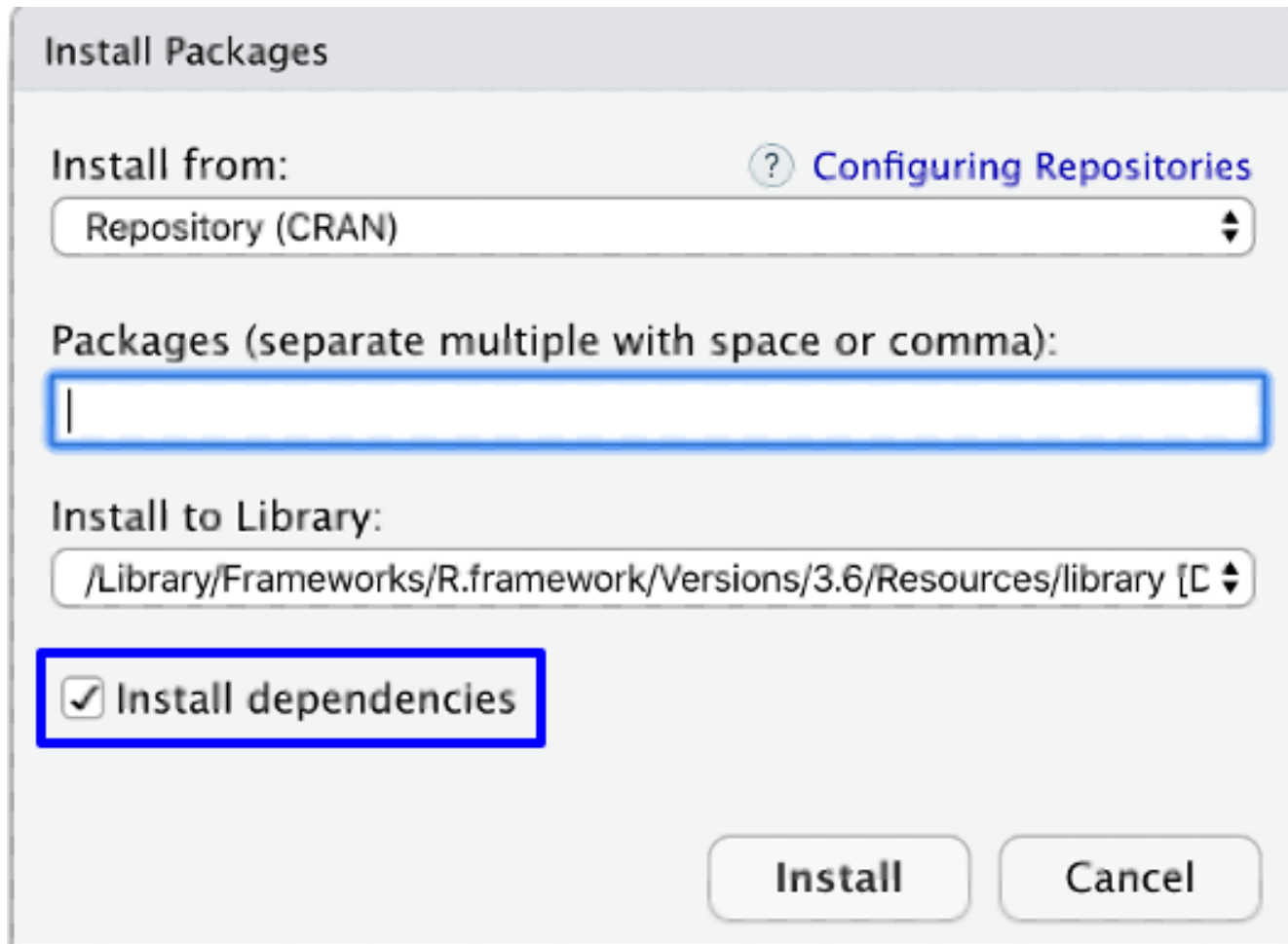
Install to Library:

/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C

☒ Install dependencies

Install Cancel

# Install Button



The image shows the 'Install Packages' dialog box in RStudio. It has a title bar 'Install Packages'. Below it, the 'Install from:' section shows a dropdown menu set to 'Repository (CRAN)' with a help icon and a link to 'Configuring Repositories'. The 'Packages (separate multiple with space or comma):' section has an empty text input field. The 'Install to Library:' section shows a text field with the default R library path. At the bottom, there is a checked checkbox for 'Install dependencies' and two buttons: 'Install' and 'Cancel'. The 'Install dependencies' checkbox and the 'Install' button are highlighted with blue rectangles.

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN)

Packages (separate multiple with space or comma):

Install to Library:

/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C]

☒ Install dependencies

Install Cancel

# R code to install packages

```
install.packages("psych")
```

# Packages

## INSTALLING

- Downloading the package and saving it to your computer
- Like installing Microsoft Word on your computer
- Do this **ONCE**

## LOADING

- Like opening Microsoft Word to write a paper
- Once a package is loaded, you can use all of it's functions and datasets are ready to use
- You need to do this **EVERY TIME** you open a new **R** session

# 2 ways to load packages

1. Checkbox in the packages tab (not recommended)
2. R code

*Either way, you need to know the name of the package*

# R code to load packages

```
library(psych)
```

# Dependencies

```
> library(lme4)  
Loading required package: Matrix  
Loading required package: Rcpp
```

Uses functions from other packages

Installed automatically

Loaded automatically

# Help! (again)

Ways to find documentation:

`?psych` -- opens documentation specific to that package or function

`??psych` -- searches for this in all documentation (that you have installed and loaded)

To find a package that does what you need: **Google**

# Summary

Packages are a collection of functions and data sets

1. You **install** the package once; must be connected to the internet
2. You **load** the package every time you use it; do not need to be connected to the internet

How do you find the function you need? How do you know what package it's in?

- **G-o-o-g-l-e!**
- "structural equation modeling in R"

How do you know how to use the function? What are the function's arguments?

- Help documentation in R
- **? function.name**